Lab 0: Machine Learning and PyCaret

BMED365-2027: Computational Imaging, Modeling and AI in Biomedicine

BMED365

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Overview

1. Introduction to Machine Learning
2. Classification vs. Regression
3. Model Evaluation
4. PyCaret
5. Notebooks and Exercises

What is Machine Learning (ML)?

Machine learning is the study of computer algorithms that improve automatically through experience.

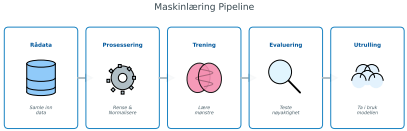
Formal definition (Tom Mitchell): A computer program is said to learn from experience **E** with respect to a task **T** and performance measure **P**, if its performance on T, as measured by P, improves with experience E.

In medicine:

- **T**: Diagnose a disease.
- **E**: Historical patient data.
- **P**: Accuracy.

Prompt

"A clean, horizontal flowchart illustrating the machine learning pipeline on white background. Steps from left to right: 1) Raw data (database icon), 2) Preprocessing (gear), 3) Training (brain-chip icon), 4) Evaluation (magnifying glass on graph), 5) Deployment (cloud). Style is modern, 'Corporate Memphis' but strictly professional and medical-technical. High quality, vector style."



ML Paradigms

- **Supervised Learning:** We have input data X and ground truth (labels) y. The goal is to learn a function $f : X \rightarrow y$.
  - *Classification:* y is a category (Sick/Healthy).
  - *Regression:* y is a continuous value (Blood pressure, Length of stay).
- **Unsupervised Learning:** We only have input data X. The goal is to find structure in the data.
  - *Clustering:* Find patient groups.
  - *Dimensionality reduction:* Simplify complex data.

Classification vs. Regression

Classification

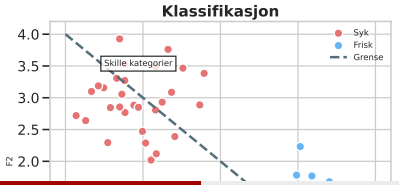
- Discrete output.
- E.g.: Diagnoses (ICD-10 codes).
- Algorithms: Logistic Regression, Decision Trees, Random Forest, SVM.

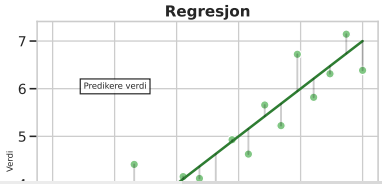
Regression

- Continuous output.
- E.g.: Time to event, dosage.
- Algorithms: Linear Regression, Random Forest Regressor, XGBoost.

Prompt

"Split screen scientific illustration. Left side labeled 'Classification': A scatter plot where a line separates red dots from blue dots. Right side labeled 'Regression': A scatter plot with a smooth curve fitting through the data points. Clean, white background, clear colors, high resolution, textbook quality."





How Do We Measure Success?

For classification we use a **Confusion Matrix**:

	Predicted Positive	**Predicted Negative**
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

- **Accuracy:** $\frac{TP+TN}{Total}$ (Can be misleading with imbalanced data!)
- **Precision:** $\frac{TP}{TP+FP}$ (How many of those we called sick are actually sick?)
- **Recall (Sensitivity):** $\frac{TP}{TP+FN}$ (How many of the sick did we find?)
- **F1-Score:** Harmonic mean of Precision and Recall.

Bias-Variance Tradeoff

A central challenge in ML is the balance between *bias* and *variance*.

Underfitting (High Bias):

- The model is too simple.
- Does not capture the pattern in the data.
- Poor on both training and test.

Overfitting (High Variance):

- The model is too complex.
- Learns "noise" in the training data.
- Good on training, poor on test.

The goal is to find the "sweet spot" that generalizes well to new data.

PyCaret: Automated ML

PyCaret is a "low-code" library that automates large parts of the ML workflow.

- **setup():** Initializes the environment, handles missing values, encodes categorical variables, splits data.
- **compare_models():** Trains and evaluates many algorithms side-by-side.
- **create_model():** Trains a specific model.
- **tune_model():** Optimizes hyperparameters automatically.
- **plot_model():** Generates standardized figures (ROC, Feature Importance).

Example PyCaret Code

```
from pycaret.classification import *

# 1. Setup
exp = setup(data=diabetes_df, target='Class variable', session_id=123)

# 2. Compare models
best_model = compare_models()

# 3. Analyze
evaluate_model(best_model)

# 4. Predict
predictions = predict_model(best_model, data=new_data)
```

Overview of Lab 0 Notebooks

1. **01-Simple_examples.ipynb**:
   - Manual implementation of simple ML concepts.
   - Understand decision trees and logistic regression "under the hood".
2. **02-Binary_classification.ipynb**:
   - Classify the Pima Indians Diabetes dataset.
   - Focus on data exploration (EDA) and model validation.
3. **03-PyCaret_quickguide.ipynb**:
   - Efficient ML workflow with PyCaret on the same dataset.
   - Comparison of how much time you save!

Key Learning Points

- **Data is most important:** "Garbage in, garbage out". Spend time understanding the data.
- **Validation:** Never trust training score alone. Use cross-validation or a hold-out test set.
- **Base rate:** Always check if the model beats a simple guess (e.g., "everyone is healthy").
- **Interpretability:** Sometimes a simple model (Linear Regression) is better than a complex one (Black Box) if we need to understand *why*.

Summary

Today we have laid the foundation for the rest of the course.

- We have defined ML.
- We have looked at the difference between classification and regression.
- We have introduced PyCaret as a powerful tool.

In the next lab (Lab 1) we will look at how we can use **Network Science** to find patients that are similar to each other (PSN).